

FACULTY OF BUSINESS & ECONOMIC SCIENCES
Department of Economics Nelson Mandela University
South campus and George Campus
ECC321/ECO304 TEST 1 (50 Marks)
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Instructions

Answer all questions

*Label your scripts with **Surname** and **student number**, failure to do so your script will not be marked*

Number your pages and questions accordingly

Question 1

1.1 What is $\bar{R}^2$ and what does $\bar{R}^2 = 0.50$ means? Why does one prefer to use $\bar{R}^2$ rather than R^2 ? (4)

1.2 Given the following equation:

$$\widehat{sleep} = 3,638.25 - .148 \text{ totwrk} - 11.13 \text{ educ} + 2.20 \text{ age}$$
$$(112.28) \quad (.017) \quad (5.88) \quad (1.45)$$
$$n = 706, R^2 = .113,$$

Is either *educ* or *age* individually significant at the 5% level against a two-sided alternative? Show your work. [10]

1.3 Regression analysis can be used to test whether the market efficiently uses information in valuing stocks. Let *return* be the total return from holding a firm's stock over the four-year period from the end of 1990 to the end of 1994. For 1990, let *dkr* be a firm's debt to capital ratio, let *eps* denote the earnings per share, let *netinc* denote net income, and let *salary* denote total compensation for the CEO.

Using the data in RETURN, the following equation was estimated:

$$\widehat{return} = -14.37 + .321 \text{ dkr} + .043 \text{ eps} - .0051 \text{ nentinc} + .0035 \text{ salary}$$
$$(6.89) \quad (.201) \quad (.078) \quad (.0047) \quad (.0022)$$
$$n = 142, R^2 = .0395.$$

Test whether the explanatory variables are jointly significant at the 5% level. Is any explanatory variable individually significant? [7]

1.4 Consider the following restricted and unrestricted equations.

$$\widehat{Price} = -14.47 + 0.976 \text{ assess}$$

$$(16.27) \quad (0.049)$$

$$n = 88 \quad SSR = 209\,448.99 \quad R^2 = 0.829$$

$$\widehat{Price} = -13.71 + 0.891 \text{ assess} + 0.431 \text{ lotsize} + 0.003 \text{ sqrft} + 0.563 \text{ bdrms}$$

$$(0.0022) \quad (0.233) \quad (0.0014) \quad (0.23) \quad (0.216)$$

$$n=88, \quad SSR=165\,644.51 \quad R^2=0.820$$

- (i) Carry out the F test for overall significance @5% of the unrestricted regression equation [6]
- (ii) Now, test $H_0: \beta_2=0, \beta_3=0, \beta_4=0$ @5% [6]

1.5 Consider the following equations that explain salaries of Managers in terms of annual firm's sales, return on equity (**roe**) and return on firms' stock (**ros**).

$$\widehat{Salary} = 4.32 + 0.280 \ln(\text{sales}) + 0.017 \text{ roe} + 0.00024 \text{ ros}$$

$$(0.32) \quad (0.035) \quad (0.0041) \quad (0.00054)$$

$$n = 209, R^2 = 0.283$$

- (i) Test the null hypothesis that *sales* have no effect on *salary* against the alternative those sales have a positive effect. Carry out the test at the 1% significant level. [5]
 - (ii) Test the null hypothesis that higher *ros* has no effect on salary, against alternative that *ros* has positive effect. Carry out the test at the 10% significant level. [5]
- 1.6 Consider the estimated equation which is used to study the effect of not attending class on final GPA:

$$\widehat{GPA} = 1.39 + 0.412 \text{ hsGPA} + 0.015 \text{ ACT} - 0.083 \text{ skipped}$$

$$(0.33) \quad (0.094) \quad (0.011) \quad (0.026)$$

$$n=141, R^2=0.234$$

Using the standard normal approximation, find the 95% confident interval for B_{hsGPA} . [7]

.....**Good luck**.....